

ACADEMIC EXPERIENCE

National University of Singapore

Assistant Professor, Department of Computer Science, School of Computing

Singapore

2026 - Present

University of Pennsylvania

Postdoctoral Researcher

Philadelphia, PA, USA

2023 - 2026

– Advisors: Prof. Weijie Su & Prof. Qi Long

EDUCATION

The Chinese University of Hong Kong, Shenzhen

Ph.D. in Computer and Information Engineering

Shenzhen, China

2017 - 2023

– Advisor: Prof. Zhi-Quan (Tom) Luo

– Thesis: Understanding Adversarially Robust Generalization: A Learning Theory Perspective

The Chinese University of Hong Kong

M.Sc. in Mathematics

Hong Kong SAR, China

2014 - 2015

Sun Yat-sen University

B.Econ., Mathematics Honors Program

Guangzhou, China

2010 - 2014

– Honors program with a dual curriculum in finance and mathematics

INDUSTRY EXPERIENCE

Tencent AI Lab

Research Intern

Shenzhen, China

2020 - 2022

GF Securities & Futures

Financial Engineer

Guangzhou, China

2015 - 2017

RESEARCH INTEREST

I am broadly interested in **learning theory, statistics and optimization**, with a focus on establishing foundations for **responsible** and **trustworthy** machine learning models. My recent research focuses on **mathematical and statistical topics in large language models (LLMs)**.

- **Theoretical Foundations for LLM Alignment:** Analyzing algorithmic biases in preference alignment methods [J3, P3], exploring game-theoretic approaches to alignment [J2], and investigating alignment from the perspective of social choice theory [P2, P4].
- **LLM Training and Fine-Tuning:** Establishing theoretical foundations for fine-tuning and post-training algorithms [C2, C3, C4, W1]. Addressing calibration issues [C1] in LLMs.
- **Adversarial Robustness:** Developing theoretical frameworks for understanding adversarial examples [J4, W2], robust overfitting [C8, C6], and adversarially robust generalization [J1, C5, C7].
- **Classical Learning Theory:** Studying optimization (convergence and stability of non-convex, non-smooth problems [C8, C6, W4]) and generalization (Rademacher complexity [J1, C5], PAC-Bayes [C7, W3]).

PUBLICATIONS

(α - β) denotes alphabetical order, * denotes equal contribution, and [†] denotes corresponding authors.

Journal Articles (Published or Under Revision)

- [J1] Adversarial Rademacher Complexity of Deep Neural Networks.
Jiancong Xiao, Yanbo Fan, Ruoyu Sun, and Zhi-Quan Luo.
Journal of Machine Learning Research (JMLR), 27, 1–45, 2026.
- [J2] Statistical Impossibility and Possibility of Aligning LLMs with Human Preferences: From Condorcet Paradox to Nash Equilibrium.
(α - β) Kaizhao Liu, Qi Long, Zhekun Shi, Weijie J. Su, and **Jiancong Xiao**.
Annals of Statistics (AoS), 2026.
- [J3] On the Algorithmic Bias of Aligning Large Language Models with RLHF: Preference Collapse and Matching Regularization.
Jiancong Xiao, Ziniu Li, Xingyu Xie, Emily Getzen, Cong Fang, Qi Long, and Weijie J. Su.
Journal of the American Statistical Association (JASA), 120, 2154–2164, 2025.
- [J4] Understanding Adversarial Robustness Against On-Manifold Adversarial Examples.
Jiancong Xiao, Liusha Yang, Yanbo Fan, Jue Wang, and Zhi-Quan Luo.
Pattern Recognition, 159, 111071, 2025.

Conference Papers

- [C1] Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach.
Jiancong Xiao^{*}, Bojian Hou^{*}, Zhanliang Wang^{*}, Ruochen Jin, Qi Long, Weijie J. Su, and Li Shen.
The 42nd International Conference on Machine Learning (ICML), 68364–68390, 2025.
- [C2] Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic.
Ruochen Jin^{*}, Bojian Hou^{*}, **Jiancong Xiao**^{*}, Weijie J. Su, and Li Shen.
The Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [C3] Preserving Diversity in Supervised Fine-tuning of Large Language Models.
Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Zhi-Quan Luo, and Ruoyu Sun.
The Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [C4] Magnetic Mirror Descent Self-play Preference Optimization.
Mingzhi Wang, Chengdong Ma, Qizhi Chen, Linjian Meng, Yang Han, **Jiancong Xiao**, Zhaowei Zhang, Jing Huo, Weijie J. Su, and Yaodong Yang.
The Thirteenth International Conference on Learning Representations (ICLR), 2025.
- [C5] Bridging the Gap: Rademacher Complexity in Robust and Standard Generalization.
Jiancong Xiao, Ruoyu Sun, Qi Long, and Weijie J. Su.
The Thirty Seventh Conference on Learning Theory (COLT), 5074–5075, 2024.
- [C6] Uniformly Stable Algorithms for Adversarial Training and Beyond.
Jiancong Xiao^{*†}, Jiawei Zhang^{*†}, Zhi-Quan Luo, and Asuman E. Ozdaglar.
The 41st International Conference on Machine Learning (ICML), 54319–54340, 2024.
- [C7] PAC-Bayesian Spectrally-Normalized Bounds for Adversarially Robust Generalization.
Jiancong Xiao, Ruoyu Sun, and Zhi-Quan Luo.
Advances in Neural Information Processing Systems (NeurIPS), 36, 36305–36323, 2023.
- [C8] Stability Analysis and Generalization Bounds of Adversarial Training.
Jiancong Xiao, Yanbo Fan, Ruoyu Sun, Jue Wang, and Zhi-Quan Luo.
Advances in Neural Information Processing Systems (NeurIPS Spotlight), 35, 15446–15459, 2022.

Preprints & Working Papers

- [P1] QuantumChem-200K: A Large-Scale Open Organic Molecular Dataset for Quantum-Chemistry Property Screening and Language Model Benchmarking.
Yinqi Zeng, Shangding Gu, **Jiancong Xiao**, and Renjie Li.
Submitted, under review, 2026.
- [P2] Theoretical Tensions in RLHF: Reconciling Empirical Success with Inconsistencies in Social Choice Theory.
Jiancong Xiao, Zhekun Shi, Kaizhao Liu, Qi Long, and Weijie J. Su.
Submitted, under review, 2025.
- [P3] A Fine-Grained Analysis of Pure Semantic Preference Alignment in Large Language Models.
Ruochen Jin^{*}, **Jiancong Xiao**^{*}, Nikhil Ruia, Qi Long, and Weijie J. Su.
Submitted, under review, 2025.

- [P4] Fundamental Limits of Game-Theoretic LLM Alignment: Smith Consistency and Preference Matching. Zhekun Shi, Kaizhao Liu, Qi Long, Weijie J. Su, and **Jiancong Xiao**[†].
Submitted, under review, 2025.

Workshop Papers

- [W1] Entropic Distribution Matching in Supervised Fine-tuning of LLMs: Less Overfitting and Better Diversity. Ziniu Li, Congliang Chen, Tian Xu, Zeyu Qin, **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo.
NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability (Best Paper Runner-up), 2024.
- [W2] Improving Adversarial Training for Multiple Perturbations through the Lens of Uniform Stability. **Jiancong Xiao**, Zeyu Qin, Yanbo Fan, Baoyuan Wu, Jue Wang, and Zhi-Quan Luo.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023.
- [W3] PAC-Bayesian Adversarially Robust Generalization Bounds for Deep Neural Networks. **Jiancong Xiao**, Ruoyu Sun, and Zhi-Quan Luo.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023.
- [W4] Smoothed-SGDmax: A Stability-Inspired Algorithm to Improve Adversarial Generalization. **Jiancong Xiao**, Jiawei Zhang, Zhi-Quan Luo, and Asuman E. Ozdaglar.
NeurIPS ML Safety Workshop, 2022.

GRANTS & FUNDING

- Contributor to NIH RO1 Grant Proposal, co-written with Prof. Weijie Su & Prof. Qi Long (2024).
 - Assisted in proposal writing, structuring key research components.

TALKS AND PRESENTATION

Invited Talks

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|---|---|
| • Statistical Foundations of LLM Alignment | Federal Reserve Bank of Kansas City, 2026.05 |
| • Statistical Foundations of LLM Alignment | Shenzhen Loop Area Institute, 2026.02 |
| • LLM Training and Coding for Statistical Learning and Data Science | American Statistical Association, 2025.11 |
| • Statistical Limits of LLM Alignment | Dartmouth College, 2025.10 |
| • Statistical Limits of LLM Alignment | ICSA 2025, University of Connecticut, 2025.06 |
| • Algorithmic Bias of RLHF | Siam MDS 2024, Atlanta, 2024.10 |
| • Algorithmic Bias of RLHF | JSM 2024, Portland, 2024.08 |
| • Adversarial Rademacher Complexity | COLT 2024, University of Alberta, 2024.06 |
| • Stability Analysis of Adversarial Training | IOS 2024, Rice University, 2024.03 |
| • Adversarially Robust Generalization | IOS 2024, Rice University, 2024.03 |
| • Algorithmic Bias of RLHF | University of Pennsylvania, 2023.11 |
| • Uniformly Stable Algorithms of Adversarial Training | MOPTA 2023, Lehigh University, 2023.08 |
| • Stability Analysis of Adversarial Training | Sichuan University, 2023.02 |
| • Adversarial Rademacher Complexity | Sichuan University, 2023.02 |
| • Stability Analysis of Adversarial Training | AI Times Forum, 2022.10 |
| • Metric Entropy | RL Seminar, CUHK(SZ), 2020.08 |

Poster Presentation

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| • Restoring Calibration for Aligned Large Language Models | ICML 2025, 2025.07 |
| • Algorithmic Bias of RLHF | 4th Penn Conference on Big Data, University of Pennsylvania, 2024.09 |
| • Algorithmic Bias of RLHF | ASSET Center AI Research Mixer, University of Pennsylvania, 2024.09 |
| • Uniformly Stable Algorithms for Adversarial Training | ICML 2024, 2024.07 |
| • Algorithmic Bias of RLHF | RL workshop, COLT 2024, University of Alberta, 2024.06 |
| • Adversarially Robust Generalization | NeurIPS 2023, New Orleans, 2023.12 |
| • Adaptive Smoothness-Weighted Adversarial Training | AdvML, ICML 2023, Hawaii, 2023.07 |

- Adversarially Robust Generalization
- Uniformly Stable Algorithms for Adversarial Training
- Stability Analysis of Adversarial Training

AdvML, ICML 2023, Hawaii, 2023.07
ML Safety, NeurIPS 2022, New Orleans, 2022.12
NeurIPS 2022, New Orleans, 2022.12

AWARDS AND SCHOLARSHIPS

- NeurIPS 2024 FITML Best Paper Runner-up 2024
- NeurIPS 2023 Scholar Award 2023
- NeurIPS 2022 Spotlight 2022
- Tencent Ph.D. Scholarship 2020 - 2023
- Poster Award of 2021 Graduate Research Conference at CUHK(SZ) 2021
- Postgraduate Scholarship by The Chinese University of Hong Kong, Shenzhen 2017 - 2023
- Undergraduate Scholarship by Sun Yat-sen University 2013

TEACHING

Instructor IVADO, Université de Montréal

- Statistical Foundations of Large Language Models Summer 2026

Instructor ICSA 2025, University of Connecticut

- Statistical Inference in Large Language Models Summer 2025

Invited Guest Lecturer University of Pennsylvania

- STAT 9917: Statistical Topics in Large Language Models Spring 2025

Teaching Assistant CUHK(SZ)

- MAT 3220: Optimization II Spring 2019 & Spring 2020
- MAT 3007: Optimization I Fall 2019
- MAT 4001: Numerical Analysis Fall 2018
- MAT 1004: Mathematical Analysis II Spring 2018
- MAT 1003: Mathematical Analysis I Fall 2017

PROFESSIONAL SERVICE

Session Organizer & Chair

- 2024 First Macau International Conference on Business Intelligence and Analytics University of Macau, 2024.12
 - Session: Optimization and Statistics for Large Language Models
- 37th Annual Conference on Learning Theory (COLT 2024) University of Alberta, 2024.06
 - Session: Adversarial and Robust Learning
- Modeling and Optimization: Theory and Applications (MOPTA 2023) Lehigh University, 2023.08
 - Session: Optimization Algorithms of Adversarial Training

Journal Reviewer

- Journal of Machine Learning Research (JMLR).
- IEEE Transactions on Information Theory (TIT).
- Journal of the American Statistical Association (JASA).
- ACM Transactions on Intelligent Systems and Technology (TIST).
- Machine Learning.
- IEEE Transactions on Audio, Speech and Language Processing (TASL).
- Quarterly Journal of Economics and Management (QJEM).

Conference Reviewer

- Machine Learning: NeurIPS 2022-2026, ICML 2023-2025, ICLR 2023-2026, AISTATS 2022 & 2026.
- Computer Vision: ECCV 2024, CVPR 2023, ICCV 2021.

LANGUAGES

- English: Fluent
- Chinese Mandarin and Cantonese: Native

REFEREES

Professor Weijie J. Su

Professor

Department of Statistics and Data Science

The Wharton School

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265 South 37th Street, Philadelphia, PA 19104

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Professor Qi Long

Professor and Vice Chair

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423 Guardian Drive, Philadelphia, PA 19104

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Professor Zhi-Quan (Tom) Luo

X.Q. Deng Presidential Chair Professor

Vice President (Academic)

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2001 Longxiang Boulevard,

Longgang District, Shenzhen 518172, China

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Associate Professor

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Longgang District, Shenzhen 518172, China

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